Supplemental Instruction Lesson Plan

Session Date: 1 FEB 2022
Course: CSCI160 (Computer Science I)

SI Leader: Miguel Maldonado
Instructor: Ken Whitener

Lesson: How do I ask a good question?

Background: Asking good questions is a cornerstone of every developer's skill set. Asking good questions will allow you to break down any problem into bite size, solvable, pieces. It will also help you to more effectively communicate with your fellow developers and greatly increase the probability that someone will respond to your question.

Practical Applications:

- Problem solving (programming)
- Communication (soft skills)
- Getting better search results
- Getting help at the workplace, on Discord, or other online forums

Agenda:

1. Problem Solving/Web Search Demo
2. Steps to Asking a Good Question
3. Posting Questions on Discord
4. Effective Communication Activity
5. Discussion/AAR

Problem Solving/Web Search Demo:

```
14
15
16 age = input(how old are you)
17

SyntaxError: invalid syntax (<string>, line 16)
```

Figure 1: Code produced a SyntaxError (using PythonTutor.com)

1. When you are unsure of the right question to ask, try searching online for key terms such as "SyntaxError: invalid syntax"
2. Research web results - https://realpython.com/invalid-syntax-python/
3. Form a well-researched question and post it to the TC3 Discord channel if you are still unsure
4. Consider using different tools to get additional information about your error

```
[miguel@framework ~]$ python
Python 3.10.2 (main, Jan 17 2022, 00:00:00) [GCC 11.2.1 20211203 (Red H
Type "help", "copyright", "credits" or "license" for more information.
>>> age = input(how old are you)
  File "<stdin>", line 1
    age = input(how old are you)
                ^^^^^^^
SyntaxError: invalid syntax. Perhaps you forgot a comma?
>>>
```

Figure 2: Code produced a SyntaxError with a suggestion (using a local instance of Python)

Steps to Asking a Good Question (from a Popular Online Coding Forum):

1) Search and Research
    - Have you thoroughly searched for an answer before asking your question? Sharing your research helps everyone. Tell us what you found and why it didn't meet your needs. This demonstrates that you've taken the time to try to help yourself, it saves energy from reiterating obvious answers, and above all, it helps you get a more specific and relevant answer!
2) Be On-Topic
    - Stick to the topic of the forum and avoid open-ended discussion.
3) Be Specific
    - Most likely, if you ask a vague question, you'll get a vague answer. But if you give details and context, it will result in a more useful answer.
4) Make it Relevant to Others
    - Make it clear how your question is relevant to more people than just you, and more people will be interested in your question and willing to investigate it.
5) Keep an Open Mind
    - The answer to your question may not always be the one you wanted, but that doesn't mean it is wrong. A conclusive answer isn't always possible. When in doubt, ask people to cite their sources, or to explain how/where they learned something. Even if they don't agree with you, or tell you exactly what you want to hear, remember: they are just trying to help.

Posting Questions on Discord:

1. Form your topic question (short version of your question). Form a question that is short, and to-the-point so that users scrolling thorugh the channel can find the right question they may be looking for. For example:

    Bad**:** C# Math Confusion
    Good**:** Why does using (float) instead of (int) yield different results when all of my inputs are integers?
    Bad**:** [python] SyntaxError
    Good**:** How do you use the input function in Python3?
2. Send your question as a message in the appropriate channel:
    - Use the general channel under the **CSCI160-CS1** group if you want to direct the question to your classmates. Usually use this if your question mostly relates to zybooks issues, or issues that is relevant only to CS1.
    - Use the **Programming Help** channel groups if you would like to ask questions to students outside of CS1 who may have knowledge with Python (or other platform) in general.

3. Right-click (control-click on mac) the message, and select **Create Thread**. Set the Thread Name to the short version of your question that you decided on in Step 1.

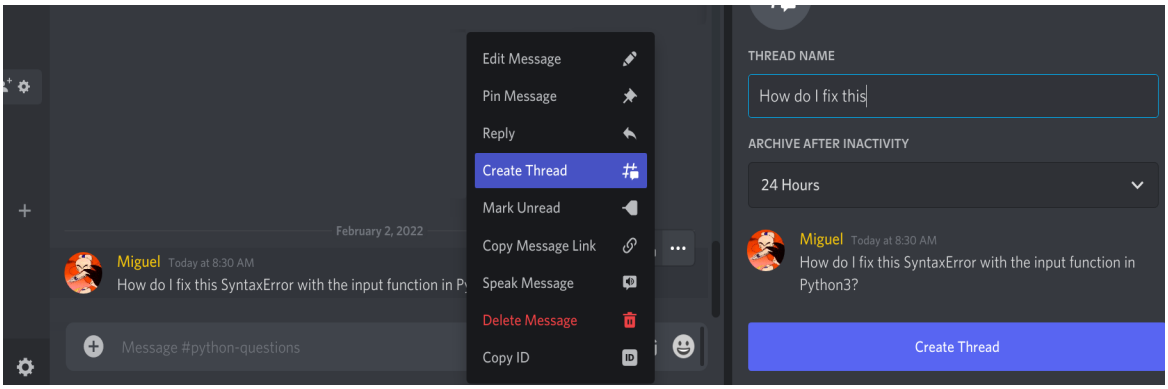


Figure 3: Creating a thread from a message

4. Provide additional details, context and your research inside your thread, so that any post concerning your question is organized and not lost to the rapid stream of "general" channels.

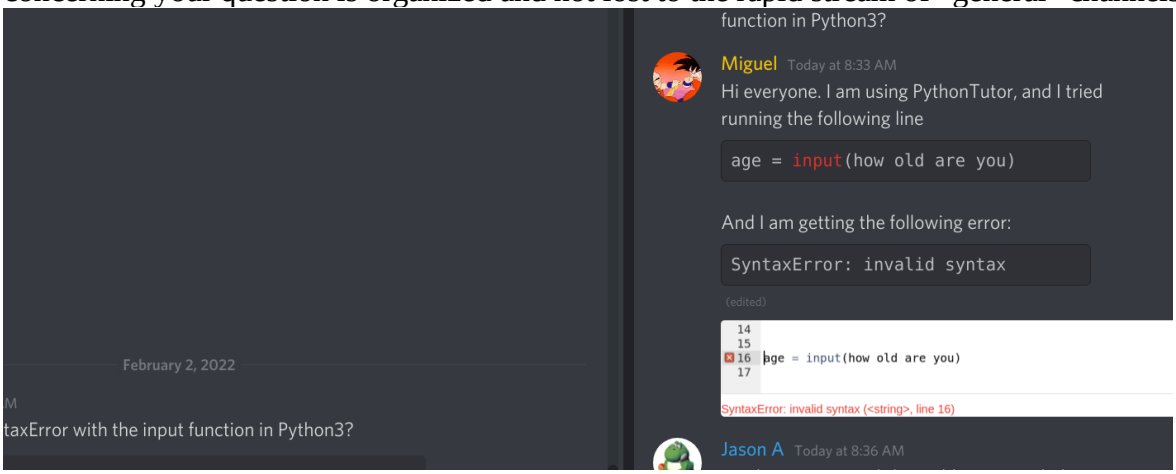


Figure 4: Adding additional information in the thread

<u>Effective Communication Activity:</u>

Steps:

1. Form pairs
2. Discuss with your partner and decide to answer one of the questions below
3. Post your answersin the CSCI 160 Discord (Remember to respond to the correct thread)
   - How do I get Discord running?
   - Where do I go to download the Discord Installer for a PC with Windows 10/11
   - Why can't I log into Discord?
   - How do I start a voice chat in Discord?
   - How do I join a server in Discord?
   - (Come up with your own)

<u>AAR/Discussion</u>

<u>Resources:</u>

- https://meta.stackexchange.com/help/how-to-ask
- https://meta.stackexchange.com/help/no-one-answers